

ASSIGNMENT REPORT-4

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For The Subject

Pattern Recognition and Machine Learning

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Introduction

A Gaussian distribution is constructed by mean vector and covariance matrix. The mean determines the centre of the distribution and the covariance matrix determines its spread. Here we constructed isotropic covariance, diagonal covariance and full covariance. Main objective is to create 2 dimensional gaussian dataset, finding the covariance, eigen vectors, eigen values and plotting its constant density curves.

Formulas

The following formulas were used in the construction of the two dimensional Gaussian distributions.

1. Multivariate Gaussian probability density function:

$$f(x) = 1 / [(2\pi)^{(d/2)} |\Sigma|^{(1/2)}] \cdot \exp[-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)]$$

2. Mean vector and covariance matrix:

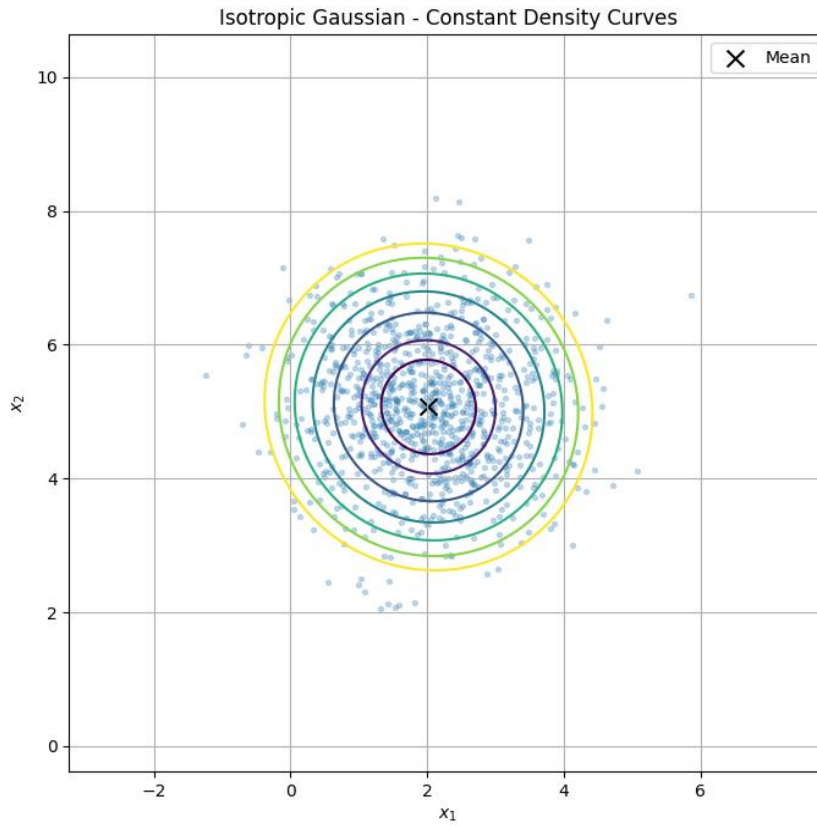
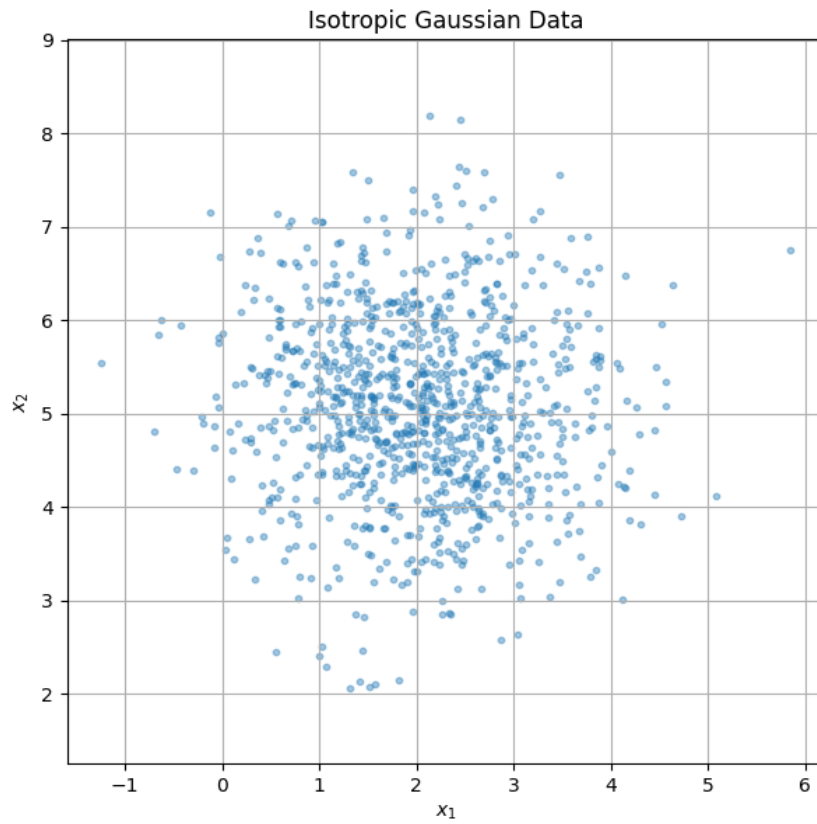
$$\mu = [\mu_1, \mu_2]^T \quad \Sigma = [[\sigma_1^2, \sigma_{12}], [\sigma_{12}, \sigma_2^2]]$$

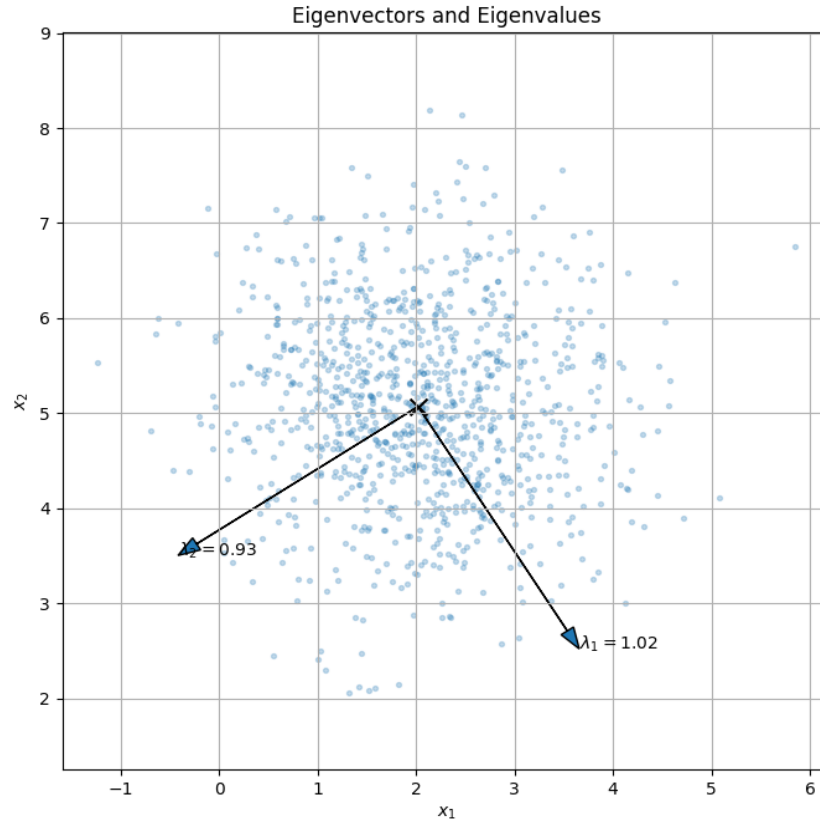
3. Constant density:

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = c^2$$

Experiments Performed

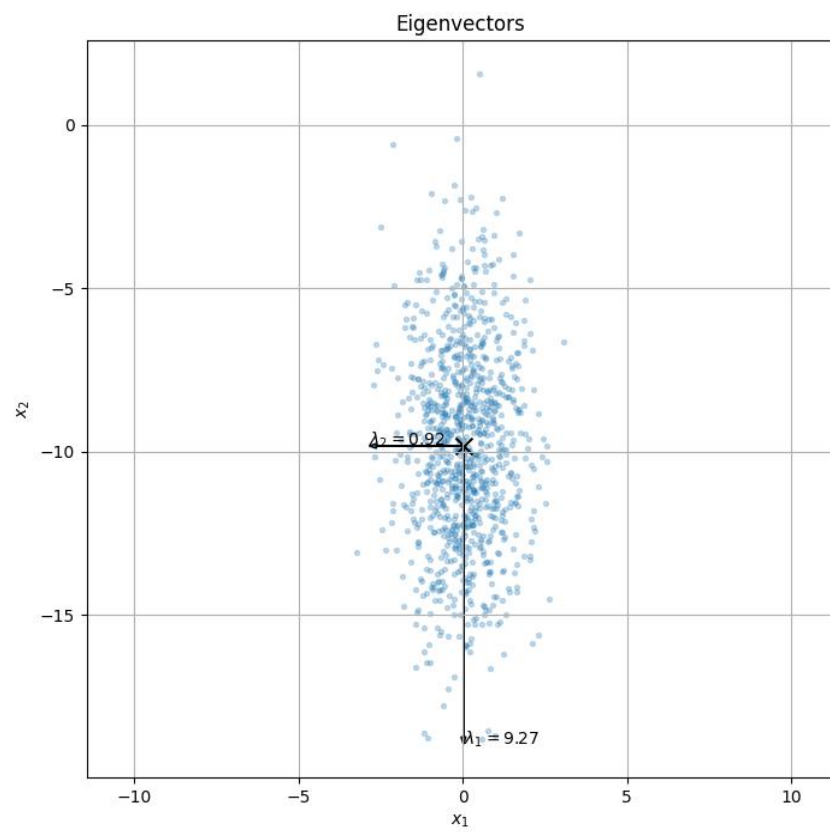
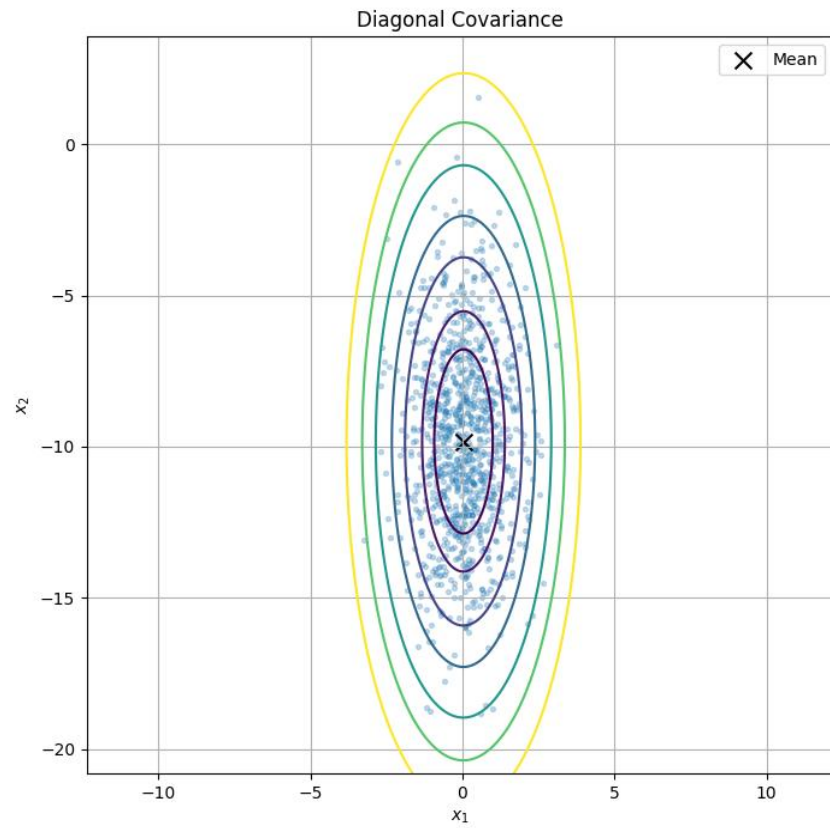
Two independent one dimensional Gaussian dataset were combined to form two dimensional observations. In isotropic, covariance matrix, eigenvalues, and eigenvectors were estimated and visualized using constant density curves.

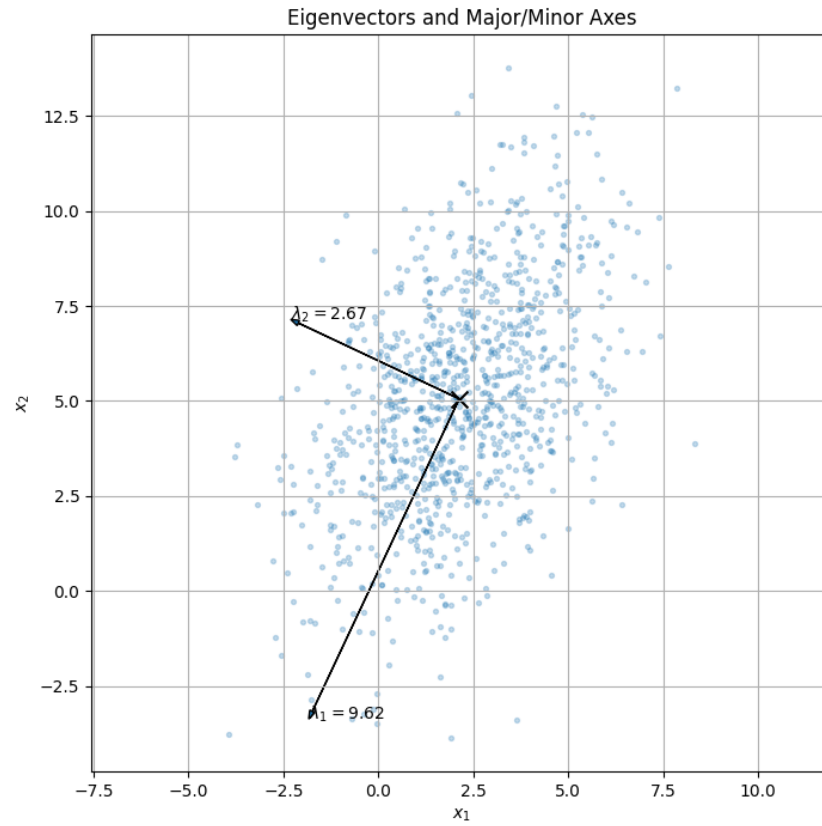
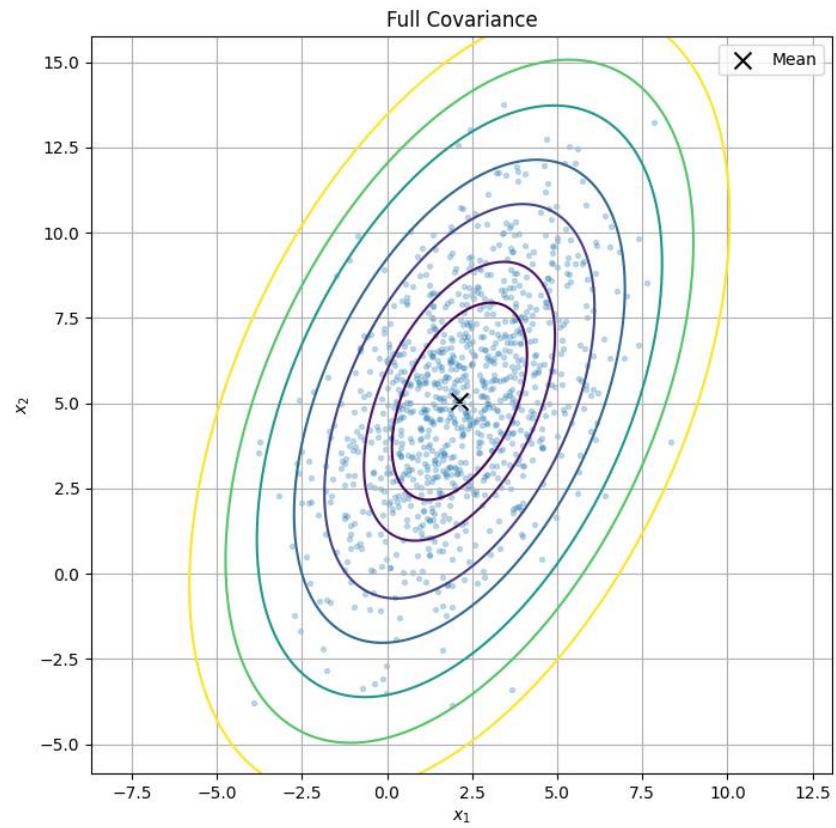




Second, a diagonal covariance matrix with different variances along the two dimensions were used to transform the generated data using $Y = \mu + \Sigma^{1/2}(X - \mu)$. And covariance, eigenvalues, eigenvectors, and constant density curves were analyzed.

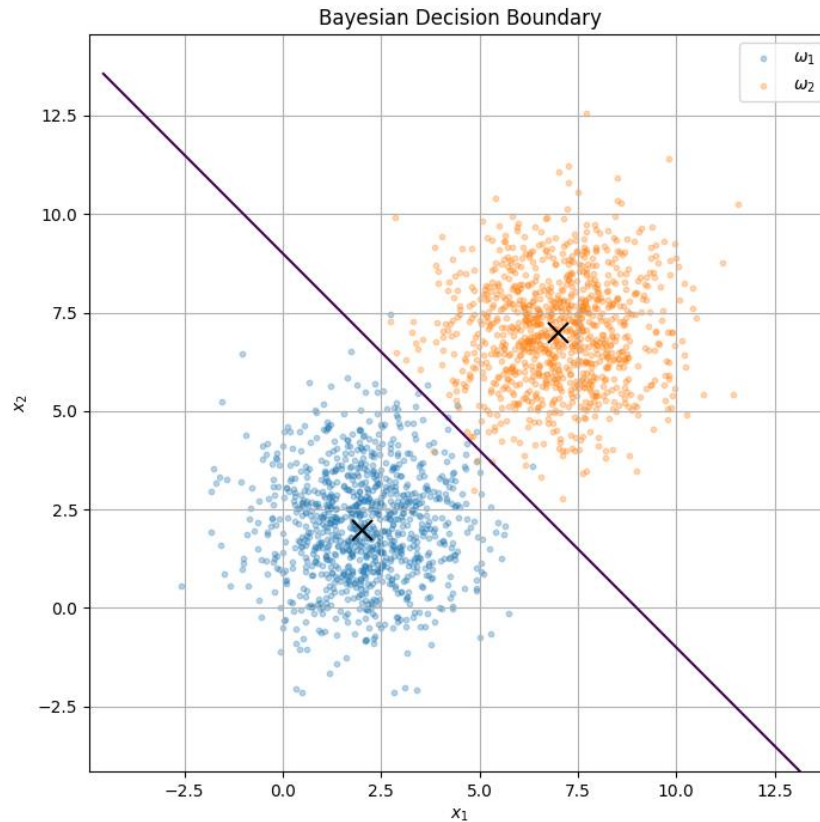
Third a full covariance matrix containing non zero off diagonal elements were used to compare between the dimensions. And the resulting orientation of the density contours and eigenvectors were studied.





Finally, two dataset representing classes ω_1 and ω_2 , each were modelled using a different Gaussian distribution. The Gaussian likelihoods were compared to visualize the classification regions and decision boundary. Different means and covariance matrices were considered to observe their effect on the boundary.





Results

The experiments demonstrated the relationship between the covariance matrix, eigenvalues and eigenvectors and also the geometry of the Gaussian density. For isotropic covariance eigenvalues were approximately equal and the constant density curves are circular because the variance is same in both directions. In diagonal covariance matrix, different variances produce elliptical contours.

For a full covariance matrix, the non zero off diagonal covariance makes elliptical contours to rotate. The eigenvectors indicate the principal directions of the ellipse, while the eigenvalues determine the scale of its principal axes. In the two class experiment the decision boundary is obtained where the posterior probabilities of the two Gaussian classes are equal. When the covariance matrices differ, the boundary can become nonlinear/quadratic, showing how both the mean and covariance matrices influence Gaussian classification.